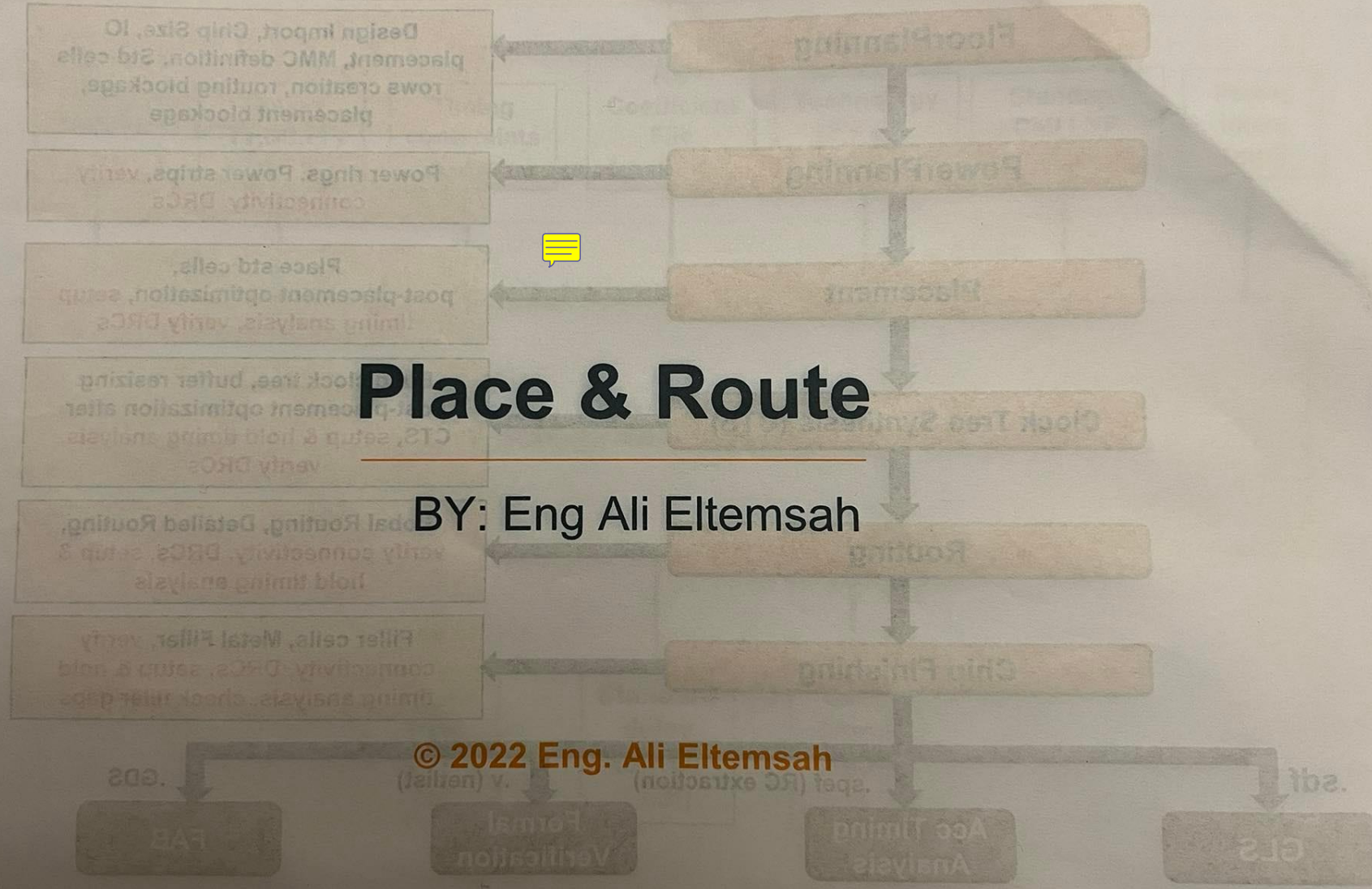


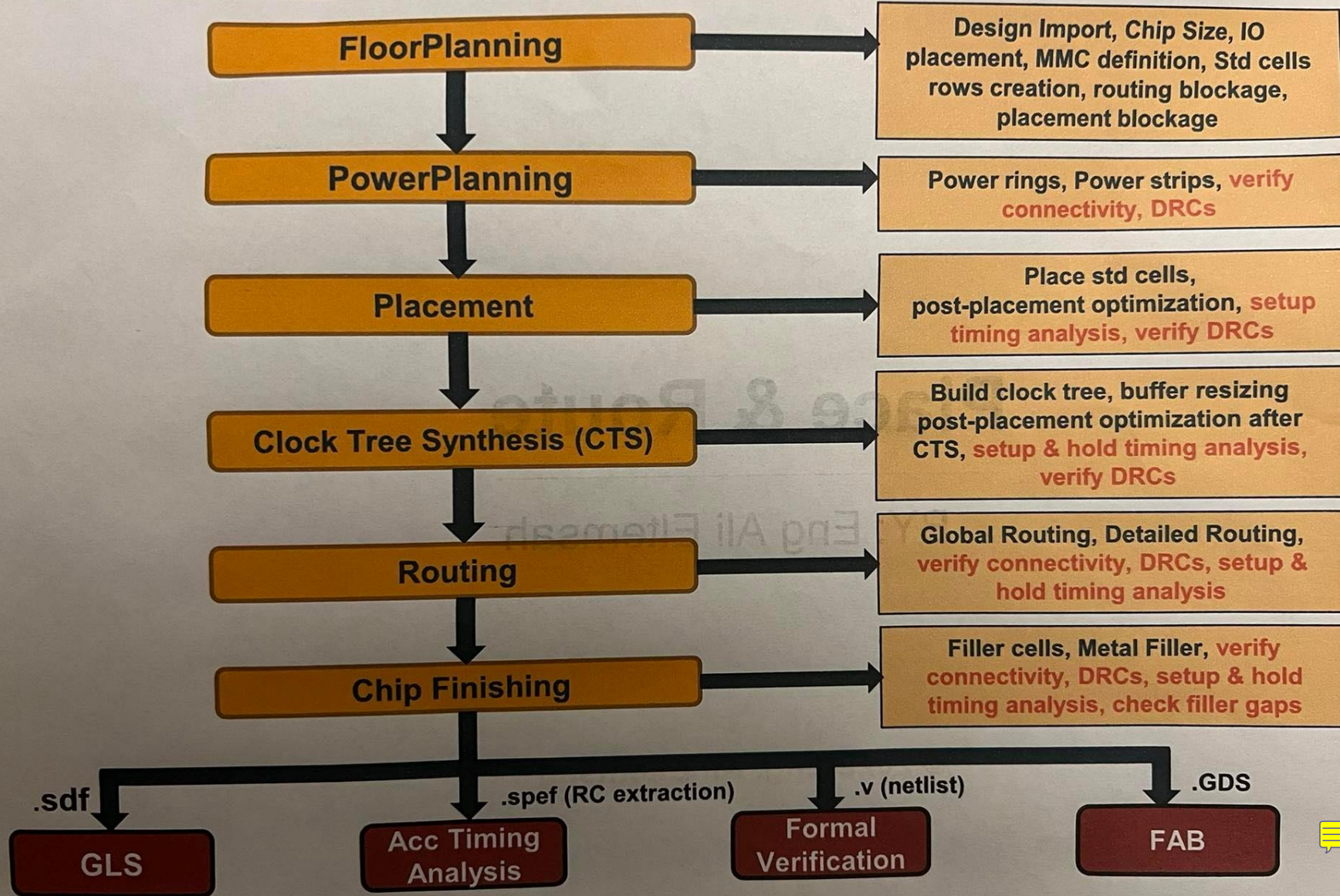
Place & Route

BY: Eng Ali Eltemsah

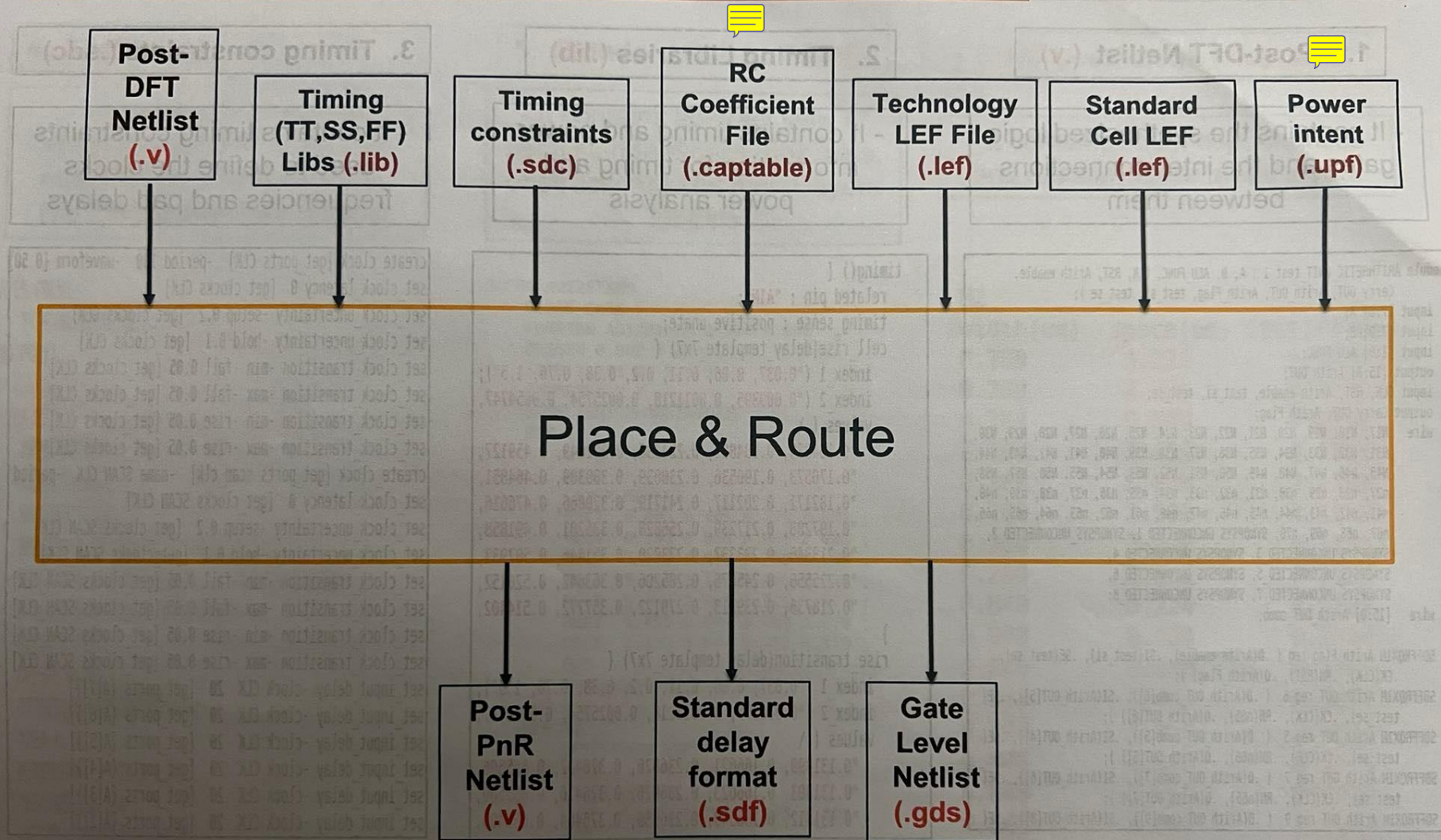
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ASIC Flow



Introduction to P&R Flow





Input Files

1. Post-DFT Netlist (.v)

- It contains the synthesized logic gates and the inter-connections between them

```
module ARITHMETIC_UNIT test1 ( A, B, ALU_FUNC, CLK, RST, Arith_enable,
    Carry_OUT, Arith_OUT, Arith_Flag, test_si, test_se );
    input [7:0] A;
    input [7:0] B;
    input [1:0] ALU_FUNC;
    output [15:0] Arith_OUT;
    input CLK, RST, Arith_enable, test_si, test_se;
    output Carry_OUT, Arith_Flag;
    wire N17, N18, N19, N20, N21, N22, N23, N24, N25, N26, N27, N28, N29, N30,
        N31, N32, N33, N34, N35, N36, N37, N38, N39, N40, N41, N42, N43, N44,
        N45, N46, N47, N48, N49, N50, N51, N52, N53, N54, N55, N56, N57, N58,
        N27, N28, N29, N30, N31, N32, N33, N34, N35, N36, N37, N38, N39, N40,
        N41, N42, N43, N44, N45, N46, N47, N48, N61, N62, N63, N64, N65, N66,
        N67, N68, N69, N70, SYNOPSYS_UNCONNECTED_1, SYNOPSYS_UNCONNECTED_2,
        SYNOPSYS_UNCONNECTED_3, SYNOPSYS_UNCONNECTED_4,
        SYNOPSYS_UNCONNECTED_5, SYNOPSYS_UNCONNECTED_6,
        SYNOPSYS_UNCONNECTED_7, SYNOPSYS_UNCONNECTED_8;
    wire [15:0] Arith_OUT_comb;

    SDFFRQX1M Arith_Flag reg ( .D(Arith_enable), .SI(test_si), .SE(test_se),
        .CK(CLK), .RN(RST), .Q(Arith_Flag) );
    SDFFRQX1M Arith_OUT_reg_6 ( .D(Arith_OUT_comb[6]), .SI(Arith_OUT[5]), .SE(
        test_se), .CK(CLK), .RN(n65), .Q(Arith_OUT[6]) );
    SDFFRQX1M Arith_OUT_reg_5 ( .D(Arith_OUT_comb[5]), .SI(Arith_OUT[4]), .SE(
        test_se), .CK(CLK), .RN(n66), .Q(Arith_OUT[5]) );
    SDFFRQX1M Arith_OUT_reg_7 ( .D(Arith_OUT_comb[7]), .SI(Arith_OUT[6]), .SE(
        test_se), .CK(CLK), .RN(n65), .Q(Arith_OUT[7]) );
    SDFFRQX1M Arith_OUT_reg_9 ( .D(Arith_OUT_comb[9]), .SI(Arith_OUT[8]), .SE(
```

2. Timing Libraries (.lib)

- It contains timing and power information for timing and power analysis

```
timing() {
    related_pin : "A1N";
    timing_sense : positive_unate;
    cell_rise(delay_template_7x7) {
        index_1 ("0.037, 0.06, 0.11, 0.2, 0.38, 0.76, 1.5");
        index_2 ("0.000395, 0.0011218, 0.0025754, 0.0054747,
            values ( \
                "0.164939, 0.184882, 0.224390, 0.302649, 0.459127,
                "0.170573, 0.190536, 0.230039, 0.308309, 0.464851,
                "0.182171, 0.202117, 0.241719, 0.320066, 0.476616,
                "0.197288, 0.217239, 0.256828, 0.335201, 0.491858,
                "0.213346, 0.233337, 0.273020, 0.351446, 0.507933,
                "0.225550, 0.245678, 0.285206, 0.363642, 0.520152,
                "0.218736, 0.239213, 0.279122, 0.357772, 0.514402,
            )
        }
    rise_transition(delay_template_7x7) {
        index_1 ("0.037, 0.06, 0.11, 0.2, 0.38, 0.76, 1.5");
        index_2 ("0.000395, 0.0011218, 0.0025754, 0.0054747,
            values ( \
                "0.131399, 0.166621, 0.236826, 0.376422, 0.655506,
                "0.131403, 0.166623, 0.236826, 0.376416, 0.655500,
                "0.131412, 0.166549, 0.236950, 0.376404, 0.655496,
```

3. Timing constraints (.sdc)

- It contains timing constraints used to define the clocks frequencies and pad delays

```
create_clock [get_ports CLK] -period 100 -waveform {0 50}
set_clock_latency 0 [get_clocks CLK]
set_clock_uncertainty -setup 0.2 [get_clocks CLK]
set_clock_uncertainty -hold 0.1 [get_clocks CLK]
set_clock_transition -min -fall 0.05 [get_clocks CLK]
set_clock_transition -max -fall 0.05 [get_clocks CLK]
set_clock_transition -min -rise 0.05 [get_clocks CLK]
set_clock_transition -max -rise 0.05 [get_clocks CLK]
create_clock [get_ports scan_clk] -name SCAN_CLK -period
set_clock_latency 0 [get_clocks SCAN_CLK]
set_clock_uncertainty -setup 0.2 [get_clocks SCAN_CLK]
set_clock_uncertainty -hold 0.1 [get_clocks SCAN_CLK]
set_clock_transition -min -fall 0.05 [get_clocks SCAN_CLK]
set_clock_transition -max -fall 0.05 [get_clocks SCAN_CLK]
set_clock_transition -min -rise 0.05 [get_clocks SCAN_CLK]
set_clock_transition -max -rise 0.05 [get_clocks SCAN_CLK]
set_input_delay -clock CLK 20 [get_ports {A[7]}]
set_input_delay -clock CLK 20 [get_ports {A[6]}]
set_input_delay -clock CLK 20 [get_ports {A[5]}]
set_input_delay -clock CLK 20 [get_ports {A[4]}]
set_input_delay -clock CLK 20 [get_ports {A[3]}]
set_input_delay -clock CLK 20 [get_ports {A[2]}]
```


Input Files

4. Technology LEF (.lef)

- It contains Technology-specific characterizations of metal layers, vias as well as Design Rule checks (DRCs)

```
LAYER POLY1
  TYPE MASTERSLICE ;
END POLY1

LAYER METAL1
  TYPE ROUTING ;
  WIDTH 0.160 ;
  SPACING 0.180 ;
  SPACING 0.18 LENGTHTHRESHOLD 1.0 ;
  SPACING 0.22 RANGE 0.3 10.0 USELENGTHTHRESHOLD ;
  SPACING 0.60 RANGE 10.05 100000.0 ;
  PITCH 0.410 ;
  OFFSET 0.205 ;
  DIRECTION HORIZONTAL ;
  AREA 0.122 ;
  MINIMUMCUT 2 WIDTH 1.40 ;
  THICKNESS 0.364 ;
  ANTENNACUMAREARATIO 5496 ;
  ANTENNACUMDIFFAREARATIO PWL ( ( 0 5496 ) ( 0.159 5496 ) )
```

5. Standard Cell LEF (.lef)

- It contains abstract view of each standard cells (box, pins, obstructions) and metal layers for pins

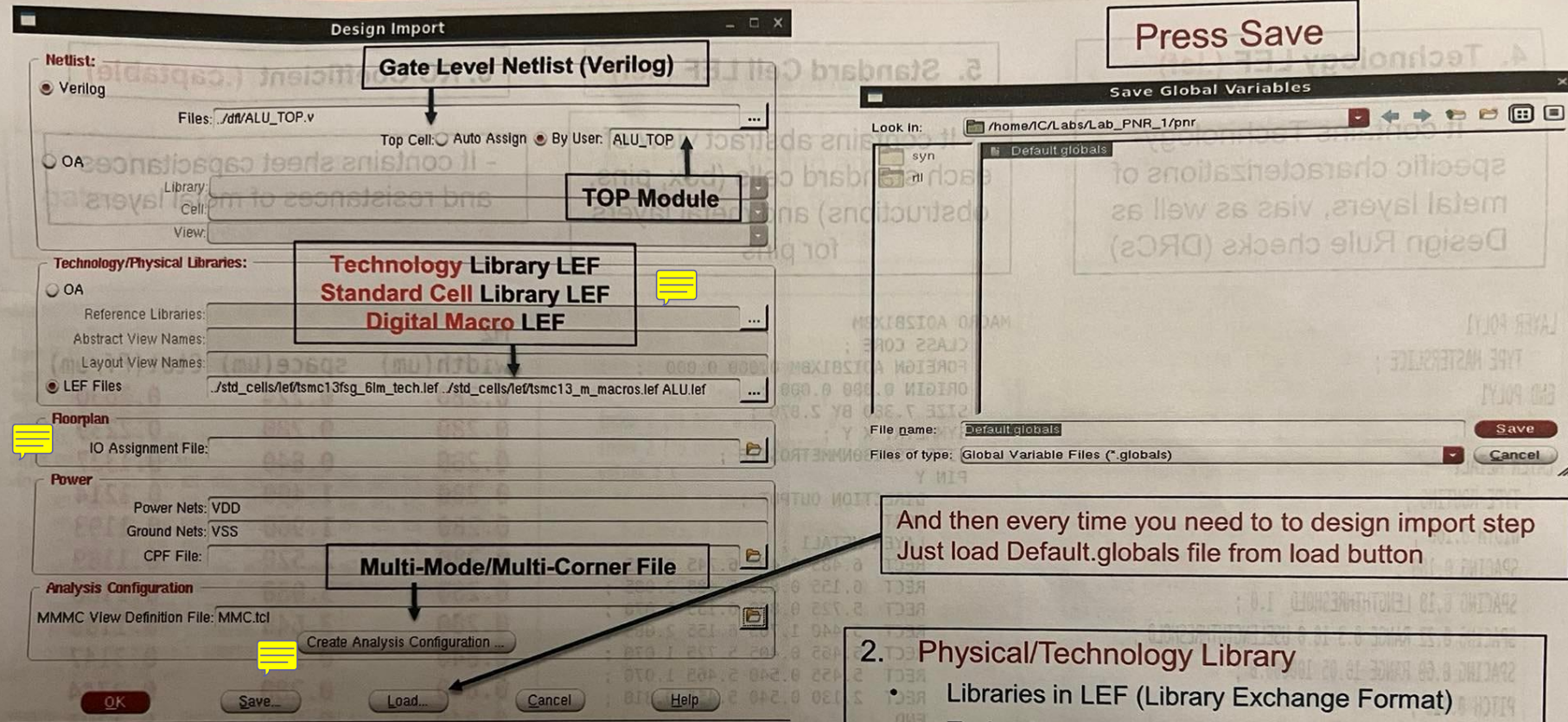
```
MACRO AOI2B1X8M
  CLASS CORE ;
  FOREIGN AOI2B1X8M 0.000 0.000 ;
  ORIGIN 0.000 0.000 ;
  SIZE 7.380 BY 2.870 ;
  SYMMETRY X Y ;
  SITE TSM130NMMETROSITE ;
  PIN Y
    DIRECTION OUTPUT ;
    PORT
      LAYER METAL1 ;
      RECT 6.485 0.405 6.745 2.085 ;
      RECT 6.155 0.800 6.485 2.085 ;
      RECT 5.725 0.800 6.155 1.070 ;
      RECT 5.440 1.765 6.155 2.085 ;
      RECT 5.465 0.405 5.725 1.070 ;
      RECT 5.455 0.540 5.465 1.070 ;
      RECT 2.130 0.540 5.455 0.810 ;
    END
    AntennaDiffArea 1.884 ;
  END Y
  PIN B0
    DIRECTION INPUT ;
    PORT
      LAYER METAL1 ;
      RECT 5.305 1.255 5.935 1.580 ;
    END
    AntennaGateArea 0.8216 ;
  END B0
```

6. RC Coefficient (.captable)

- It contains sheet capacitances and resistances of metal layers

M2	width(um)	space(um)	Ctot(Ff/um)
	0.280	0.224	0.2630
	0.280	0.280	0.2239
	0.280	0.840	0.1327
	0.280	1.400	0.1214
	0.280	1.960	0.1193
	0.280	2.520	0.1189
	0.280	3.080	0.1188
	0.280	3.640	0.1188
	0.840	0.224	0.3147
	0.840	0.280	0.2744
	0.840	0.840	0.1795
	0.840	1.400	0.1677
	0.840	1.960	0.1654
	0.840	2.520	0.1650
	0.840	3.080	0.1649
	0.840	3.640	0.1649
	2.000	0.224	0.4074

Design Import



1. Verilog gate-level netlist(s)

- Gates from the standard cell library
- Design can be hierarchical or flat

And then every time you need to to design import step
Just load Default.globals file from load button

2. Physical/Technology Library

- Libraries in LEF (Library Exchange Format)
- Technology LEF
 - characterizations of metal layers, vias, etc
- Standard Cell LEF
 - Abstract view of each cell (box, pins, obstructions)
 - Includes metal layers for pins

Design Import Files

3. Setting up MMMC (Multi-Mode/Multi-Corner) analysis

1. Create libraries set (Timing Libraries (Worst, Typical, Best))

```
create_library_set -name min_library -timing "scmetro_tsmc_cl013g_rvt_ff_1p32v_m40c.lib«
```

2. Create different **constraints Modes** (Timing constraints file from synthesis)

```
create_constraint_mode -name func_mode -sdc_files { ../syn/modulo6_1.sdc}
```

3. Create different **delay Corners** (Timing Libraries (Worst, Typical, Best) + RC Corner)

```
create_rc_corner -name RCcorner -cap_table {../captable/bicmos8hp_6AM_31_nm.CapTbl}
```

4. Create different **delay Corners** (Timing Libraries (Worst, Typical, Best) + RC Corner)

```
create_delay_corner -name min_corner -library_set min_library -rc_corner Rccorner
```

5. Create Analysis View (delay corner + constraints Mode)

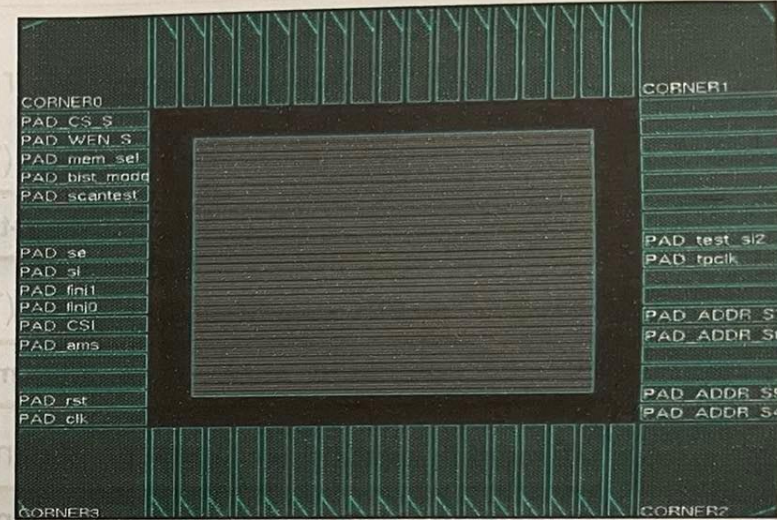
```
create_analysis_view -name setup_view -delay_corner max_corner -constraint_mode func_mode  
create_analysis_view -name hold_view -delay_corner min_corner -constraint_mode func_mode
```

6. Set analysis view to above for both setup and hold

```
set_analysis_view -setup {setup_view} -hold {hold_view}
```

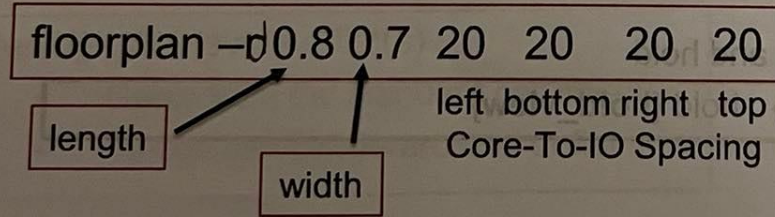

IO, P/G Placement

- Specify placement of I/O pins on the "IO box"
 - Input/output circuits (I/O Pads) are intermediate structure connecting internal signals from the core of the IC to the External pins of the chip package.
 - I/O Pads consists of some logic cells as level shifters to control voltage level and buffers to increase driving strength
 - I/O Pads contains ESD (Electrostatic Discharge) protection to protect the chip from emergency flooding of Charges which can break down the chip



Specify Floorplan

- Specify by size or by coordinates
 - Core size (aspect ratio)
 - Core utilization (leaves space for routing)
 - Core to IO boundary (leaves space for power rings)



Basic Advanced

Design Dimensions

Specify By: ☒ Size ☐ Die/IO/Core Coordinates

Core Size by: ☒ Aspect Ratio: Ratio (H/W): 344954128

Core Utilization: 0.699966

Cell Utilization: 0.699966

Dimension: Width: 19.62 Height: 14.4

Die Size by: Width: 39.62 Height: 34.4

Core Margins by: ☒ Core to IO Boundary ☐ Core to Die Boundary

Core to Left: 10.0 Core to Top: 10.0

Core to Right: 10.0 Core to Bottom: 10.0

Die Size Calculation Use: ☐ Max IO Height ☒ Min IO Height

Floorplan Origin at: ☒ Lower Left Corner ☐ Center

Unit: Micron

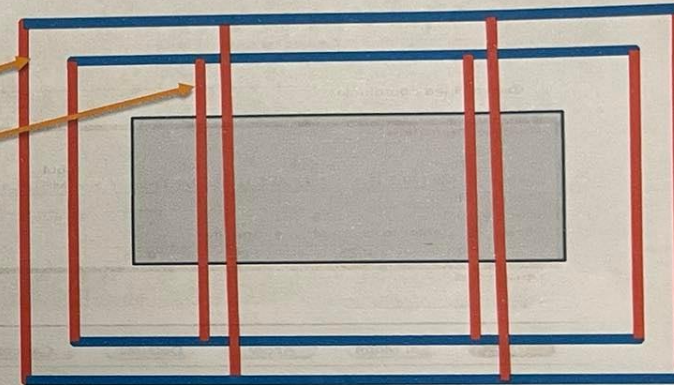
OK Apply Cancel Help

Power Planning

- Power planning means to provide power to the every macros, standard cells, and all other cells are present in the design. Power and Ground nets are usually laid out on the metal layers.
- For core logic there is a core ring enclosing the core with one or more sets of power and ground rings. The next consideration is to construct cell power and ground that is internal to core logic these are called power and ground stripes

- Level of power distribution

- **Rings:** carries VDD and VSS around the chip
- **Stripes:** carries VDD and VSS from the ring
- **Rails:** connect VDD and VSS to the standard cell's VDD and VSS



- IR Drop:

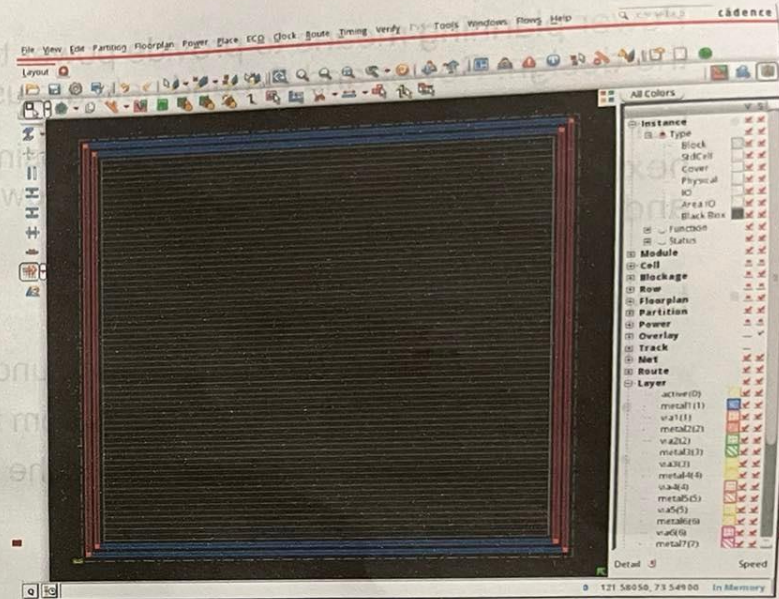
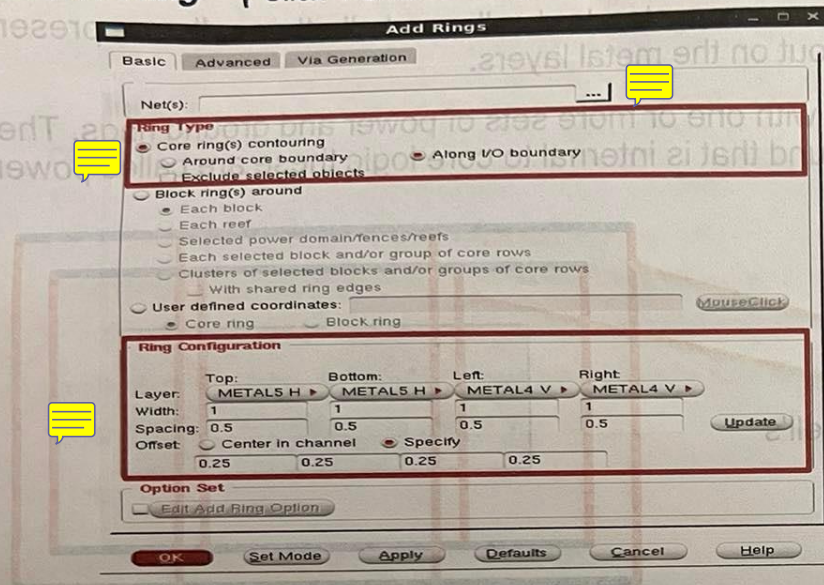
- The power supply in the chip is distributed uniformly through metal layers across the design and these metal layers have their finite amount of resistance. When we applied the voltage the current starts flowing through these metal layers and some voltage is dropped due to that resistance of a metal wire and current. This drop is called IR drop. Because of IR drop, delay will increase and it violates the timing. And this will increase noise and performance will be degraded.

- Methods to improve static IR drop

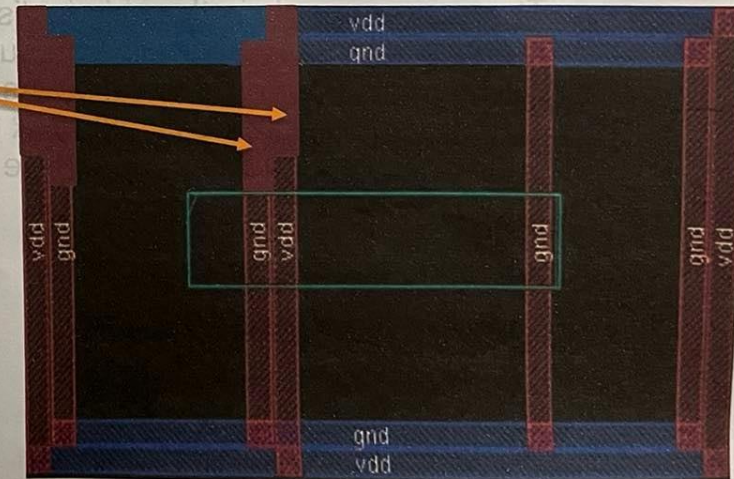
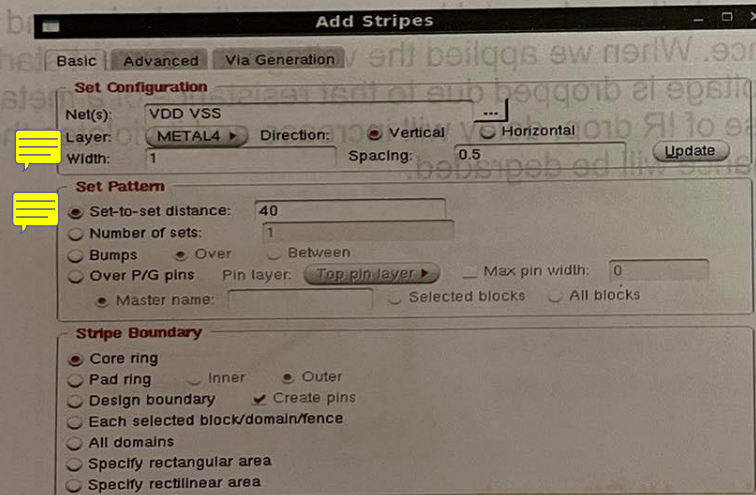
- Increase the width of the straps.
- Increase the number of wires.

Power Planning

- **Power Ring** (Click "Power" → "Power Planning" → "Add Rings...")

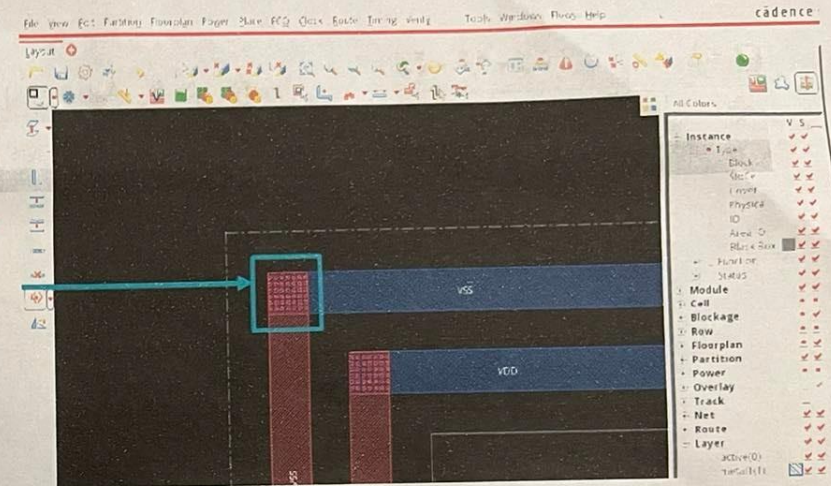
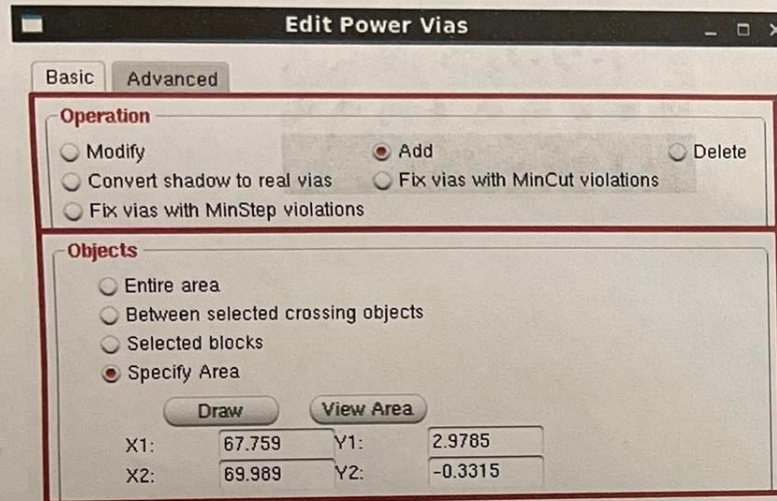


- **Power Stripes** (Click "Power" → "Power Planning" → "Add Stripes...")

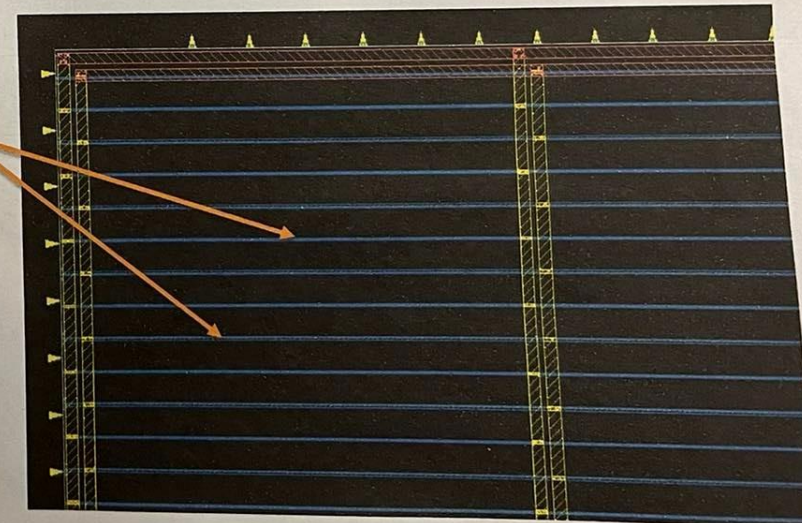
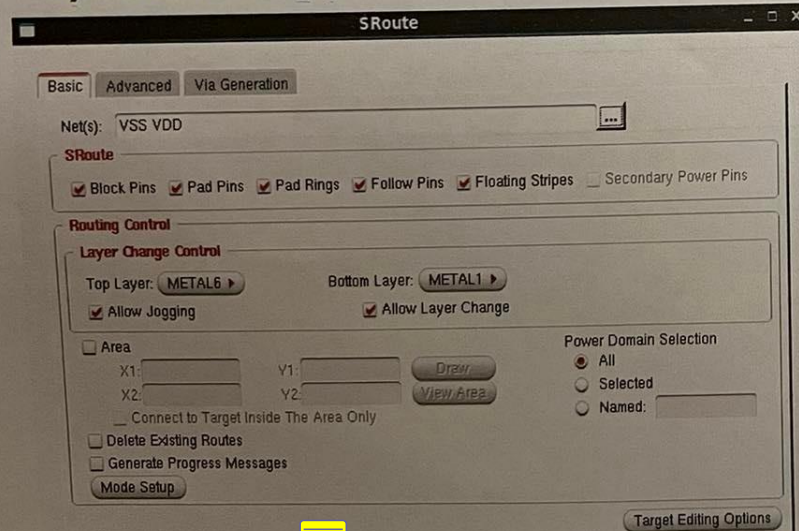


Power Planning

- **Power VIA** (Click "Power" → "Power Planning" → "Edit Power Vias...")



- **Special Routing** (Click "Route" → "Special Route...")



Tools

- **Add Ruler**



- **Remove Ruler**



- **Wire Cutter**



Shortcuts

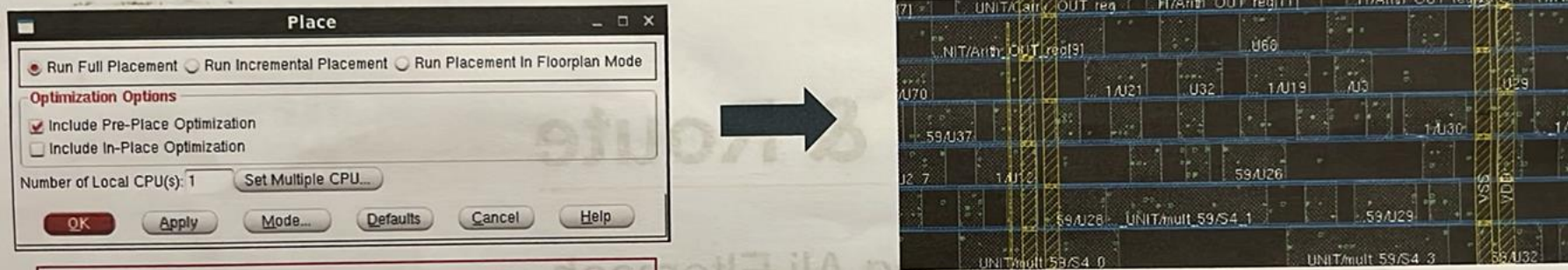
- **M** → *Move Wires*
- **S** → *Stretch Wires*
- **A** → *Idle state*
- **F** → *Full Screen*

Placement

1. Add Placement Blockage



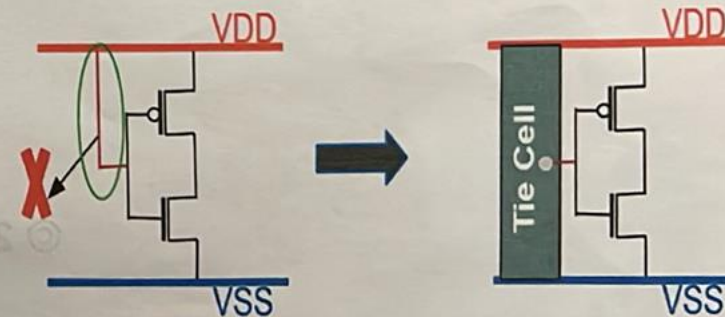
2. Place standard cells (Click “Place” → “Place Standard Cell ...” → “OK”)



placeDesign -inPlaceOpt -prePlaceOpt

3. Place Tie-Low and Tie-High Cells

- Every netlist is having any pin connected to 0 logic or 1 logic (like .A(1'b0) or .B(1'b1). If the gate of the transistor connects directly to VDD or VSS for a constant high/low input signal, and in case any glitch arises in the supply voltage it results in damage of sensitive gate oxide of the transistor. To avoid the damages mentioned above, we avoid the direct connection from VDD or VSS to the input of any logic gates. A tie cell is used to connect the input of any logic to the VDD or VSS
- Tie cells are not present in the synthesized netlist and not placed in the initial placement of the standard cells. Tie cells are inserted in the placement stage



addTieHiLo -cell TIELOM -prefix LTIE
addTieHiLo -cell TIEHIM -prefix HTIE

Timing Analysis

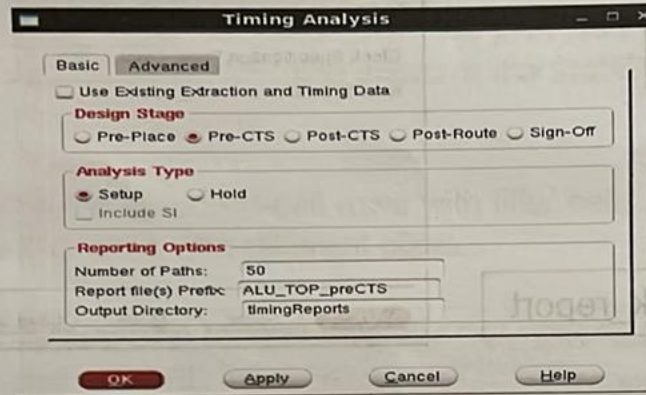
3. Run Setup/Hold Analysis (Click “Timing” → “Timing Analysis”)

Design Stage:-

- pre-CTS
- post-CTS
- Post-Route

Analysis Type :-

- setup
- hold



timeDesign Summary

Setup views included:
setup3_analysis_view

Setup mode	all	reg2reg	default
WNS (ns):	0.000	N/A	0.000
TNS (ns):	0.000	N/A	0.000
Violating Paths:	0	N/A	0
All Paths:	50	N/A	0

- “timing Reports” folder is created and contains all timing reports



Timing Optimization

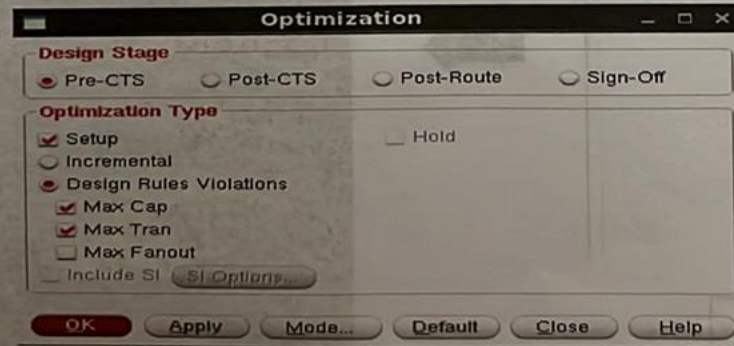
4. Optimize design for setup/hold (Click “ECO” → “Optimize Design”)

Design Stage:-

- pre-CTS
- post-CTS
- Post-Route

Analysis Type :-

- setup
- hold



timeDesign Summary

Setup views included:
setup3_analysis_view

Setup mode	all	reg2reg	default
WNS (ns):	0.000	N/A	0.000
TNS (ns):	0.000	N/A	0.000
Violating Paths:	-7	N/A	0
All Paths:	0	N/A	0

Clock Tree Synthesis

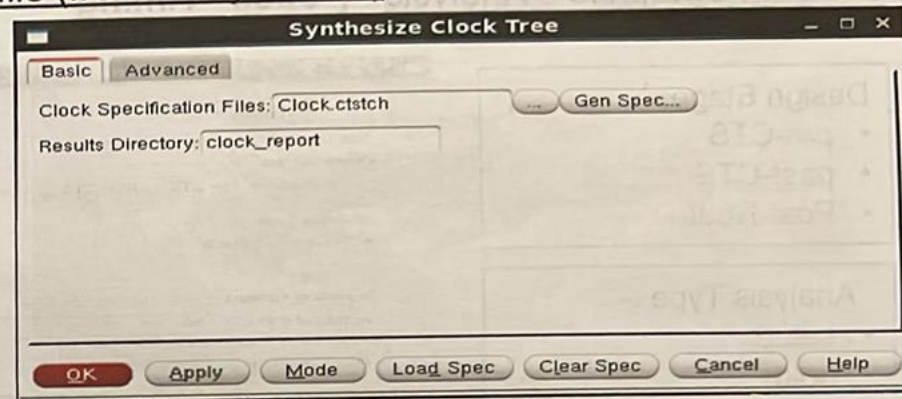
1. Create the clock tree spec file (**Clock.ctstch**) from the .sdc file (from dft synthesis)

```
setCTSMODE -specMultiMode true -engine ck  
clockDesign -genSpecOnly Clock.ctstch
```

2. Perform clock tree synthesis (CTS)

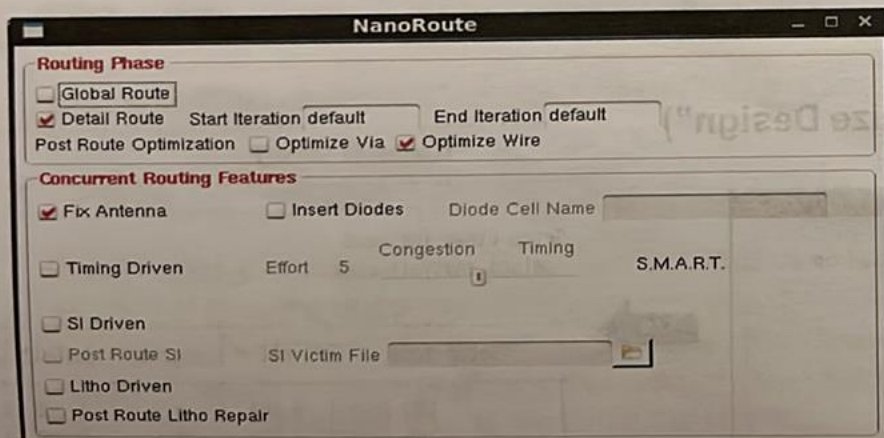
(Click “Clock” → “Synthesize Clock Tree ...”)

```
clockDesign -specFile Clock.ctstch -outDir clock_report
```



Routing

1. Create the clock tree spec file (**Clock.ctstch**) from the .sdc file (from dft synthesis)



```
routeDesign -globalDetail -viaOpt -wireOpt
```

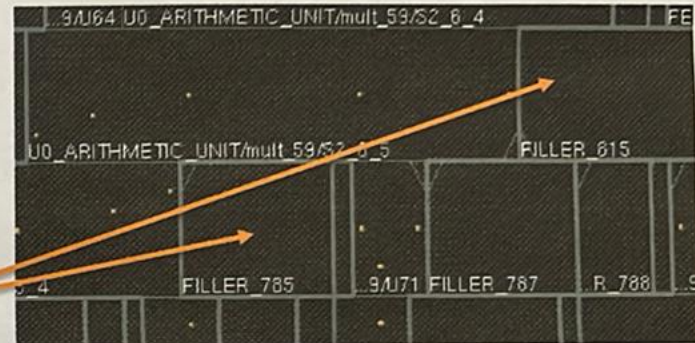


Chip Finishing

Filler Cells

- Filler cells have no logical connectivity. these cells are provided for continuity in the rows for VDD and VSS nets to ensure that all power nets are connected and improve the stability of the power supply. It also contains substrate nwell connection to improve substrate biasing.
- you can fill empty space in the standard-cell rows with filler cells. Std cells library provide filler cells with different sizes.

```
set filler_cell_list {FILL1M FILL2M FILL4M FILL8M FILL16M FILL32M FILL64M}  
addFiller -cell $filler_cell_list -prefix FILLER -markFixed
```



• Generate Output Files

1. Generate Gate Level Netlist File
 - saveNetlist export/\${top_cell}.v
2. Generate GDS File
 - streamOut export/\${top_cell}.gds -mapFile gds2InLayer -libName DesignLib -stripes 1 -units 2000 -mode ALL
3. Generate SDF File
 - delayCal -sdf export/\${top_cell}.sdf -version 3.0

• Reports

1. Power Report
 - report_power -outfile report/power.rpt
2. Area Report
 - report_area > report/area.rpt

Innovus Flow Steps

- Floorplanning

1. Design Import
2. Determine Chip size
3. I/O Assignments

- Power planning

1. Power Ring
2. Power Stripes
3. Special Routing

- Placement

1. Placement Blockage
2. Standard cell Placement
3. Tie low/high cells Placement

- CTS

1. Placement Blockage
2. Standard cell Placement
3. Tie low/high cells Placement

- Routing

1. Global/Detailed Routing

- Chip Finishing

1. Add Filler Cells
2. Generate post-PNR Netlist
3. Generate GDS File
4. Generate SDF File

- Checks

- Verify Connectivity
- Verify Geometry (DRCs)

- Checks

- Verify Connectivity
- Setup Timing Analysis

- Checks

- Verify Connectivity
- Setup Timing Analysis
- Hold Timing Analysis

- Checks

- Verify Connectivity
- Verify Geometry (DRCs)
- Setup Timing Analysis
- Hold Timing Analysis

- Checks

- Verify Connectivity
- Verify Geometry (DRCs)
- Setup Timing Analysis
- Hold Timing Analysis
- Filler Gaps